

Independent method-selection v3 holdout: source and label gate

Status (2026-09-29): **the candidate inputs have been staged and aligned, but the prespecified independent NS-AUC comparison is not runnable; no external method outcome or new model-selection API choice was scored.** Every new download and audit has a chained local trace. Source binaries remain under ignored `artifacts/`; the linked JSON audits retain hashes, counts and trace paths.

Inputs and exact alignment

The pinned [CDRPipe comparative-analysis repository](#) contributed 233 CREEDS disease-signature CSVs, CMap metadata and a gene map. The [staged-input audit](#) records 227,110/227,110 finite disease gene/logFC rows. Its public [Box release](#) provided the 243,722,737-byte CMap rank matrix (SHA-256 29e7dcbcab5af810d6fd0d21e534df231494d1c1756cd46cf6518308f68d860). The [matrix alignment audit](#) verifies 13,071 probe IDs against 6,100 drug-experiment columns, with no missing rank cells. These are ranks, not yet demonstrated to be signed perturbation log-fold changes interchangeable with TRANSCRIPT's expression vectors.

Three pinned [Open Targets Platform 26.06](#) parquets—clinical indications, diseases and drug molecules—were downloaded and SHA-256 checked in the [input receipt](#). The [exact crosswalk audit](#) uniquely maps 141/233 disease signatures by preferred name and another 47 by exact synonym; 894 CMap name/DrugBank combinations map by a unique preferred drug name. Their intersection contains 3,345 distinct recorded clinical-indication pairs over 167 disease signatures. This is **positive/development-activity evidence only**: an absent Open Targets link is unknown, not a documented failed indication. The newer 26.06 release, matching rules and denominator make this count non-comparable with CDRPipe's older published validation count.

The original [repoDB source](#) at commit `cd4edec67e57df1d2a5f552e30a6960a38d79432` supplied a 10,800-row RData table ([source audit](#), [status domains](#)). It contains 6,677 Approved and 4,123 Terminated/Withdrawn/Suspended rows, as described by the [repoDB paper](#). These latter statuses mean trial stoppage, **not necessarily biological inefficacy**; recorded reasons include funding and administrative closure. The [strict crosswalk audit](#) uses unique normalized disease-name-to-UMLS-CUI matches and unambiguous CMap DrugBank IDs, excludes conflicting approved/failed pairs, and never turns an unknown into a negative. It finds 67 mapped disease signatures, 165 approved pairs, 78 stopped-trial pairs and 14 diseases with both label types; **zero** has at least eight of each needed as a conservative disease-level five-inner-fold eligibility screen.

The [CUI overlap audit](#) finds 49 of those 67 mapped diseases already occur in the analyzed TRANSCRIPT disease matrix. Excluding those IDs leaves 18 disease signatures, only 28 approved and 12 stopped-trial pairs in total and only **two** diseases with both labels. Different CUIs would still not prove different upstream GEO/CREEDS cohorts: TRANSCRIPT's README says its disease features also came from CREEDS. Therefore the files do not establish independent evaluation units. Neither the CMap

rank orientation nor feature harmonization for B0p/B1k/B1/B2 has been frozen or validated.

A second [ontology bridge audit](#) checked Open Targets UMLS cross-references, accepting only a unique repoDB CUI per disease and still requiring unambiguous CMap DrugBank IDs. This increases the matched set to 107 signatures (347 approved, 124 stopped-trial pairs), but **zero** diseases meet the conservative eight-per-label screen. The separate [fold-minimum audit](#) tests the weaker necessary—not sufficient—condition of at least five of each label: only two signatures pass, and both overlap TRANSCRIPT, leaving **zero** source-ID-disjoint candidates even at that minimum. Removing the 71 overlapping TRANSCRIPT CUIs leaves 36 signatures (79 approved, 25 stopped-trial pairs), with only five containing both labels. The broader bridge therefore does not change the feasibility conclusion; ontology cross-references are also not proof of independent underlying expression cohorts.

Consequence for the frozen v3 question

The [v3 protocol](#) requires a source-independent benchmark, explicit positive and negative associations, identical outer splits/five-fold inner tuning and official NS-AUC for fixed B2, a deterministic selector and one prescore LLM selector choice per independent unit. The staged Open Targets labels fail the negative-label gate. The strict repoDB crosswalk fails the disease-unit/fold gate and has substantial development-set overlap. Pooling diseases, relabeling unknowns, accepting fuzzy ID matches or changing the endpoint to positive-only recall **after seeing these feasibility counts** would be a separately named exploratory study, not fulfillment of the frozen confirmatory v3 test. No API call has been made on external outcomes and no external performance claim follows from this audit.

The alternative [TRACE release](#) covers three diseases and has curated positive/negative controls, but it does not provide a sufficiently large set of independent disease-level units with the same raw drug-perturbation feature representation for this protocol. A defensible next attempt needs a larger, source-disjoint expression-and-label collection (or an explicitly approved redesigned endpoint), documented drug/disease crosswalk and fold eligibility, then a new prescore protocol freeze before any selector call or outer score.